III B.Tech II Semester

23A30603b	**PREDICTIVE ANALYTICS (Professional Elective-II)**	**L**	**T**	**P**	**C**
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Course Objectives:

- To introduce the fundamental concepts and techniques of predictive analytics.
- To apply statistical models and machine learning algorithms for prediction.
- To interpret model performance using evaluation metrics.
- To explore feature engineering, model tuning, and cross-validation.
- To implement predictive solutions for real-world business and research problems.

Course Outcomes:

Upon successful completion of the course, students will be able to:

- Understand the principles and importance of predictive analytics.
- Apply regression and classification models for predictive tasks.
- Perform data preprocessing, feature selection, and transformation.
- Evaluate and validate models using standard metrics.
- Design predictive solutions to solve domain-specific challenges.

UNIT I: Introduction to Predictive Analytics
Introduction to Predictive Analytics and Business Intelligence, Types of Predictive Models: Classification, Regression, Time Series, Supervised vs Unsupervised Learning, Predictive Modeling Workflow, Applications in Marketing, Finance, Healthcare, Challenges in Predictive Analytics.

UNIT II: Data Preparation and Feature Engineering
Data Cleaning: Handling Missing, Noisy, and Inconsistent Data, Feature Selection and Dimensionality Reduction (PCA, LDA), Feature Scaling: Normalization, Standardization, Encoding Categorical Variables, Feature Extraction and Construction, Dealing with Imbalanced Datasets.

UNIT III: Predictive Modeling with Regression and Classification
Linear Regression and Polynomial Regression, Logistic Regression for Binary Classification, Decision Trees and Random Forest, k-Nearest Neighbors (k-NN) and Naïve Bayes, Support Vector Machines (SVM), Model Selection and Comparison.

UNIT IV: Model Evaluation and Validation
Training, Testing, and Validation Sets, Cross-Validation Techniques (k-Fold, Stratified, LOOCV), Evaluation Metrics: Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix and Classification Report, Bias-Variance Trade-off and Overfitting, Hyperparameter Tuning: Grid Search, Random Search.

UNIT V: Advanced Topics and Applications
Ensemble Learning: Bagging, Boosting (AdaBoost, XGBoost), Predictive Analytics with Time Series (ARIMA, Prophet), Deep Learning for Predictive Modeling (ANNs, LSTM), Use of Predictive Analytics in IoT, Retail, and Healthcare, Ethics and Privacy in Predictive Analytics, Building and Deploying End-to-End Predictive Systems.

Textbooks:

1. **Dean Abbott,***Applied Predictive Analytics: Principles and Techniques for the Professional Data Analyst*, Wiley, 2014.
2. **John D. Kelleher, Brendan Tierney**, *Data Science: Predictive Analytics and Data Mining*, MIT Press, 2018.

Reference Books:

1. **Galit Shmueli et al.**, *Data Mining for Business Analytics: Concepts, Techniques, and Applications in R*, Wiley, 2017.
2. **Eric Siegel**, *Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die*, Wiley, 2016.
3. **Trevor Hastie, Robert Tibshirani, Jerome Friedman**, *The Elements of Statistical Learning*, Springer, 2009.

Online Learning Resources:

- https://www.coursera.org/specializations/predictive-analytics– Coursera Specialization
- https://www.edx.org/course/data-science-and-machine-learning-capstone – edX Predictive Analytics Courses
- https://www.kaggle.com/learn/intro-to-machine-learning – Kaggle Tutorials